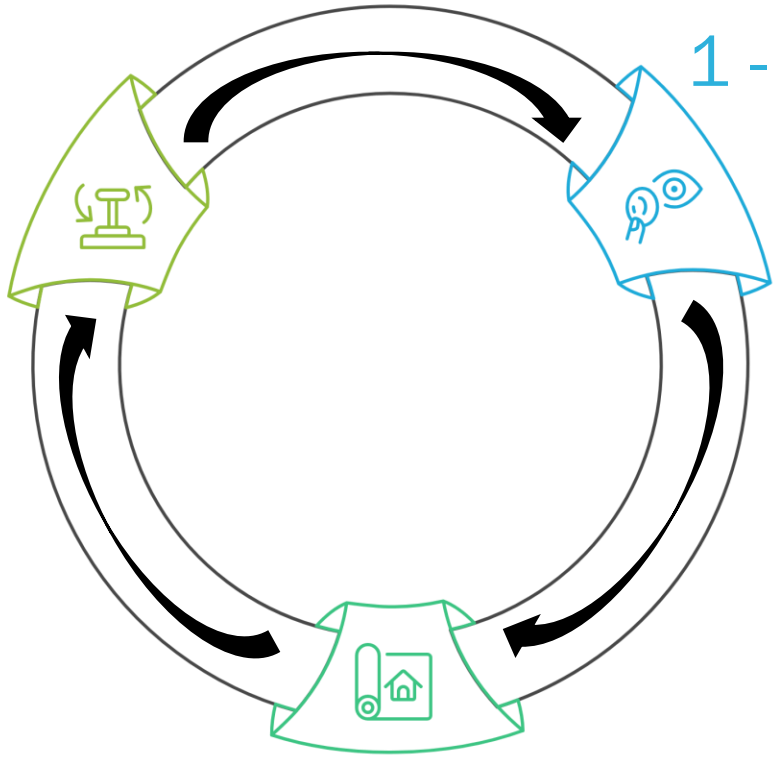
A Continual Learning Framework for Opponent Detection in Autonomous Car Races

Master's Thesis in AI Engineering

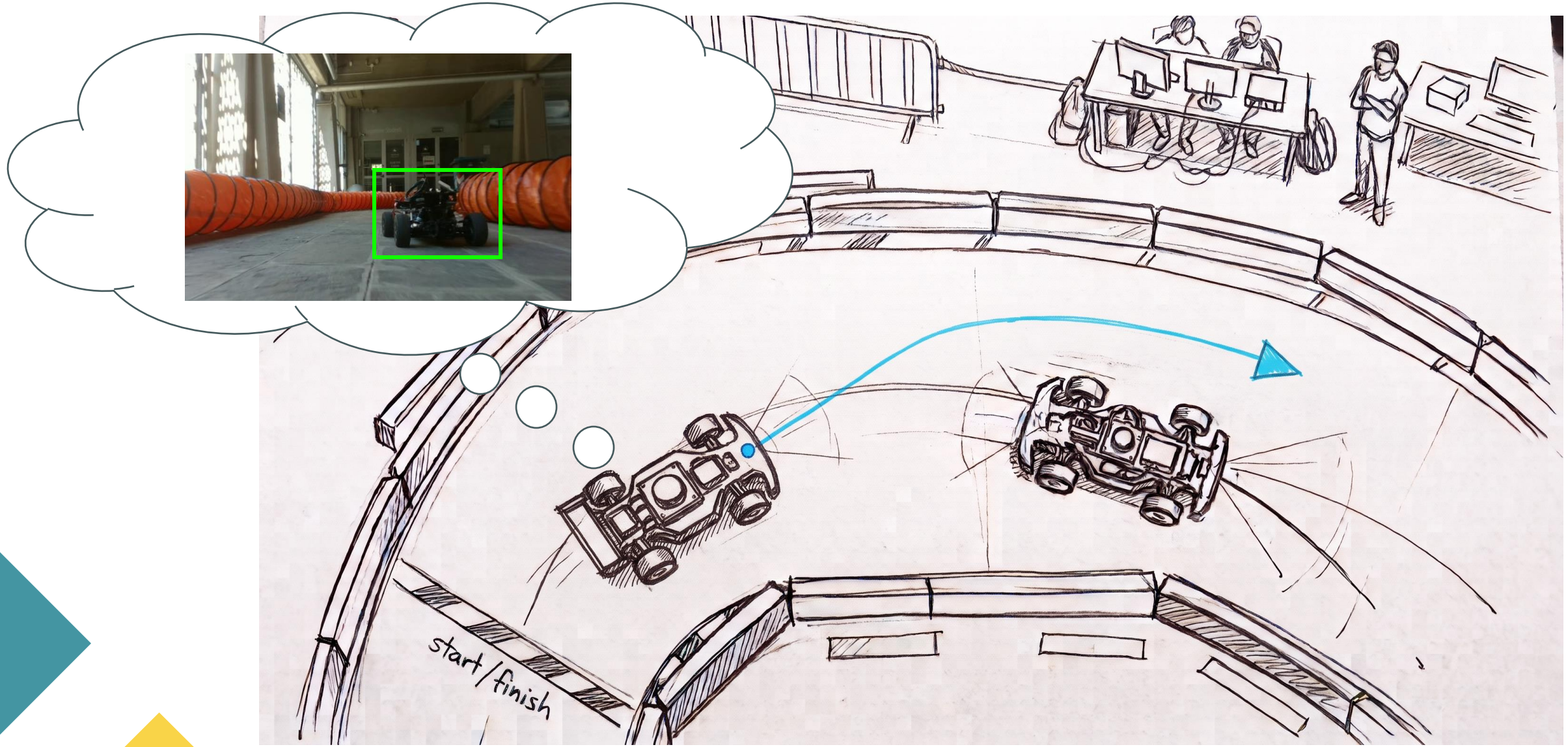
Candidate: Tommaso Mortara

Supervisor: Prof. Paolo Burgio

Co-supervisors: Prof. Alessandro Capotondi, Francesco Moretti



Autonomous Systems

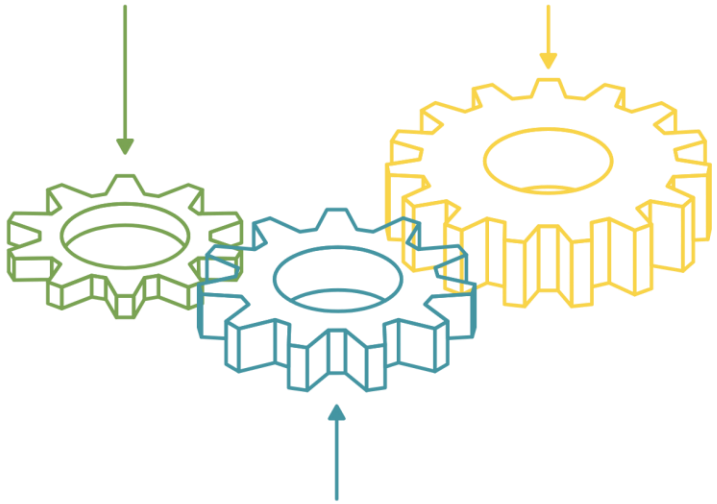


Camera-Based Detection

Why creating a dataset ex-novo?

Publicly Available Data Research

Final Decision: Custom Dataset Creation

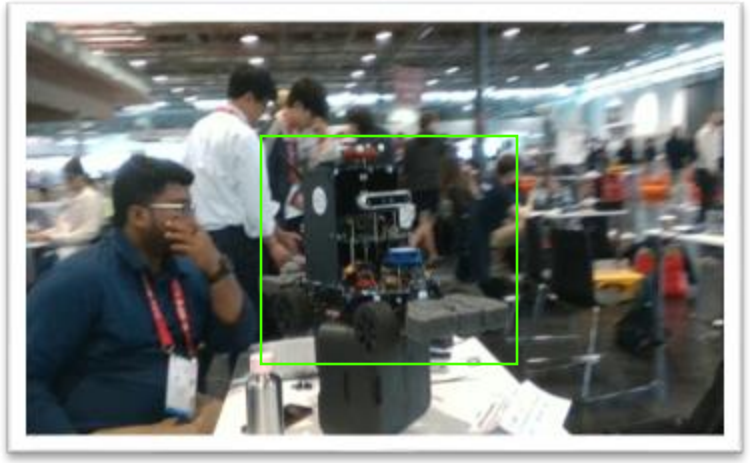


Evaluation of Research Results

Problem #1 Lack Of Data



Dataset1 sample



Dataset2 sample



Dataset3 sample

New Sessions Data Variety
+
Time Constraints



Fast Retraining to Add Knowledge



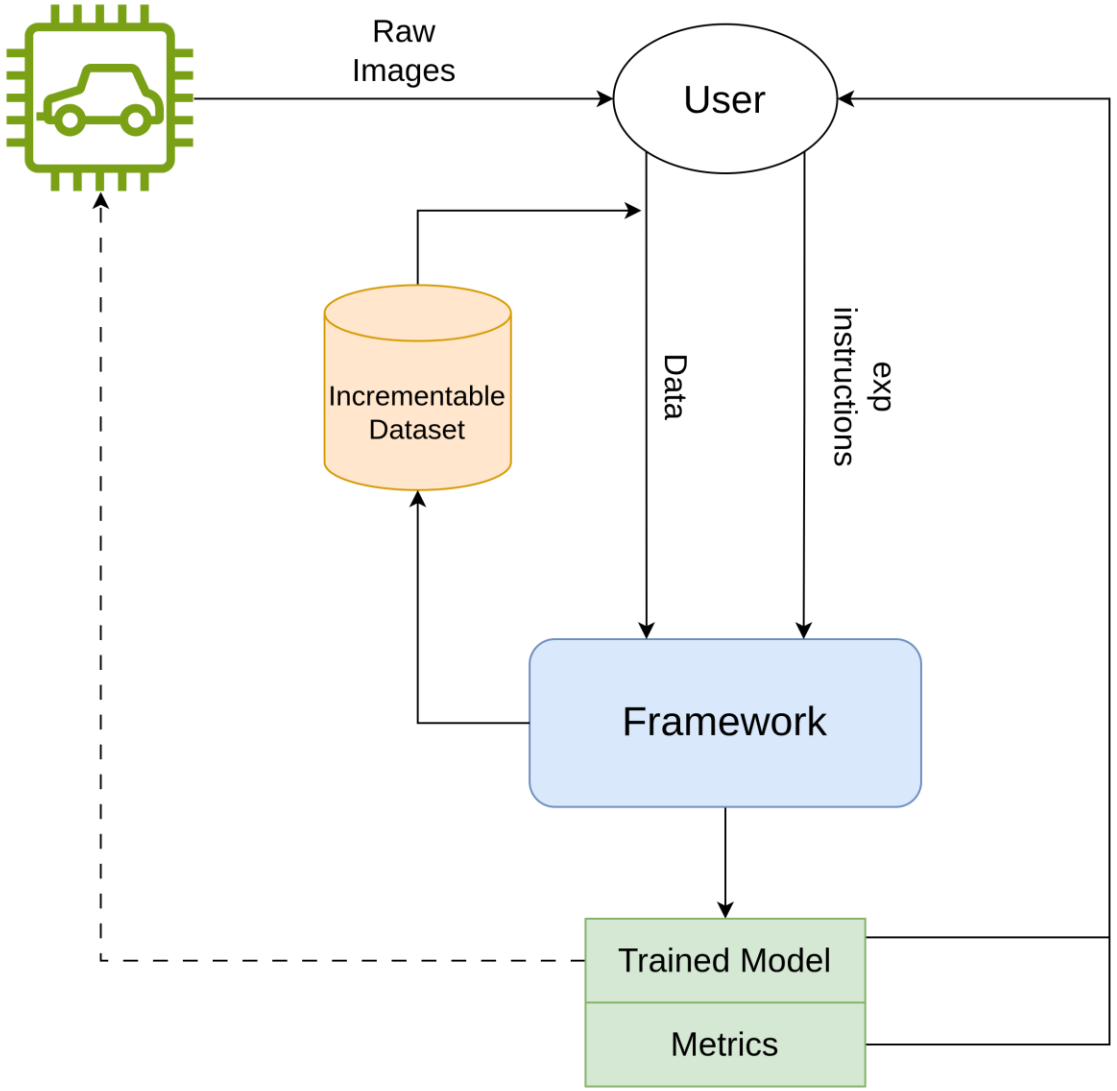
Continual Learning

Problem #2
Successive Sessions

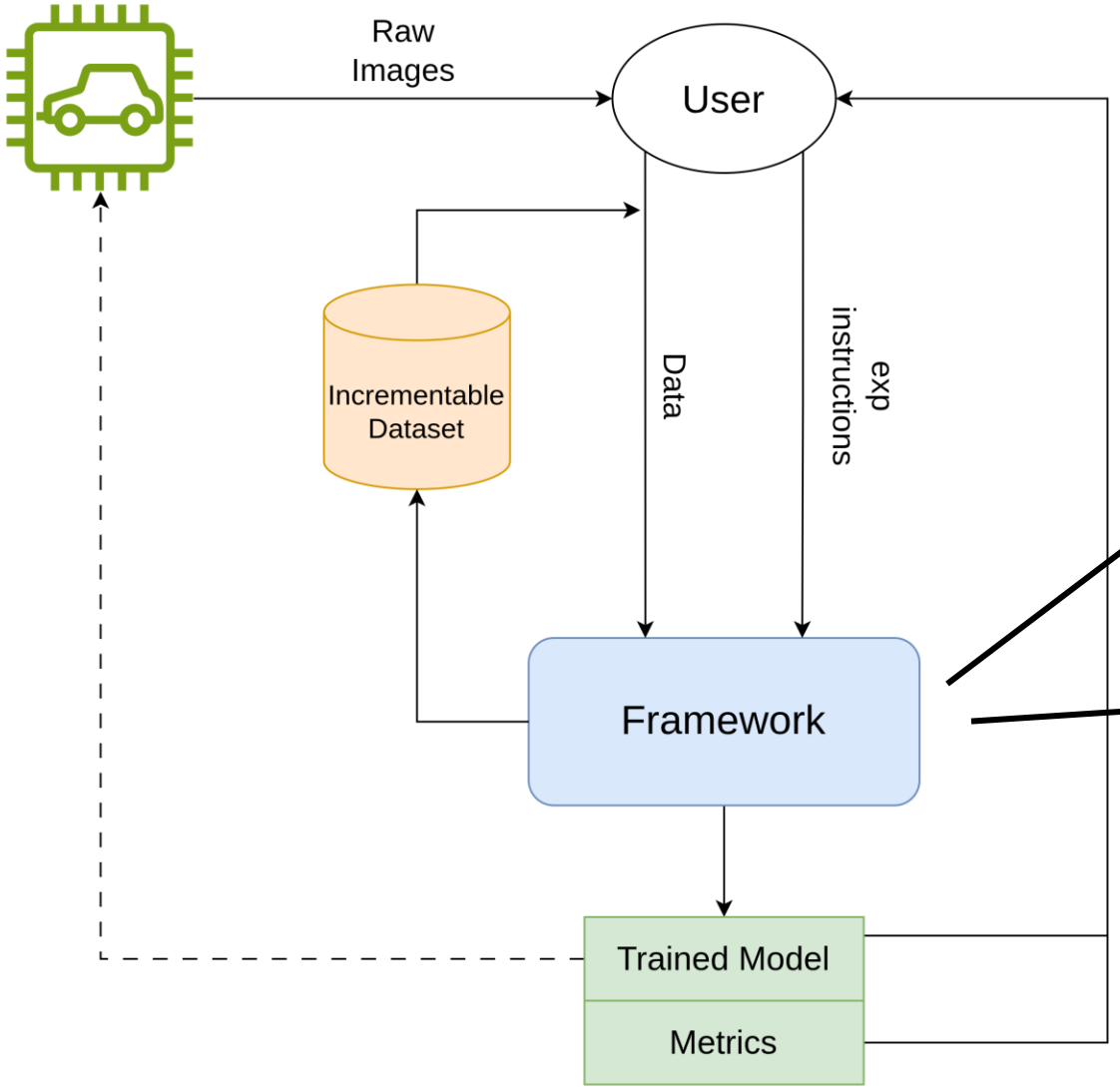
Contributions

1. Dataset well suited for incrementation
2. Framework for evaluating experiments of various nature
3. Experiments and considerations on continual learning scenario in autonomous racing

Framework


Raw
Images
User
Incrementable
Dataset
Data
exp
instructions
Framework
Trained Model
Metrics

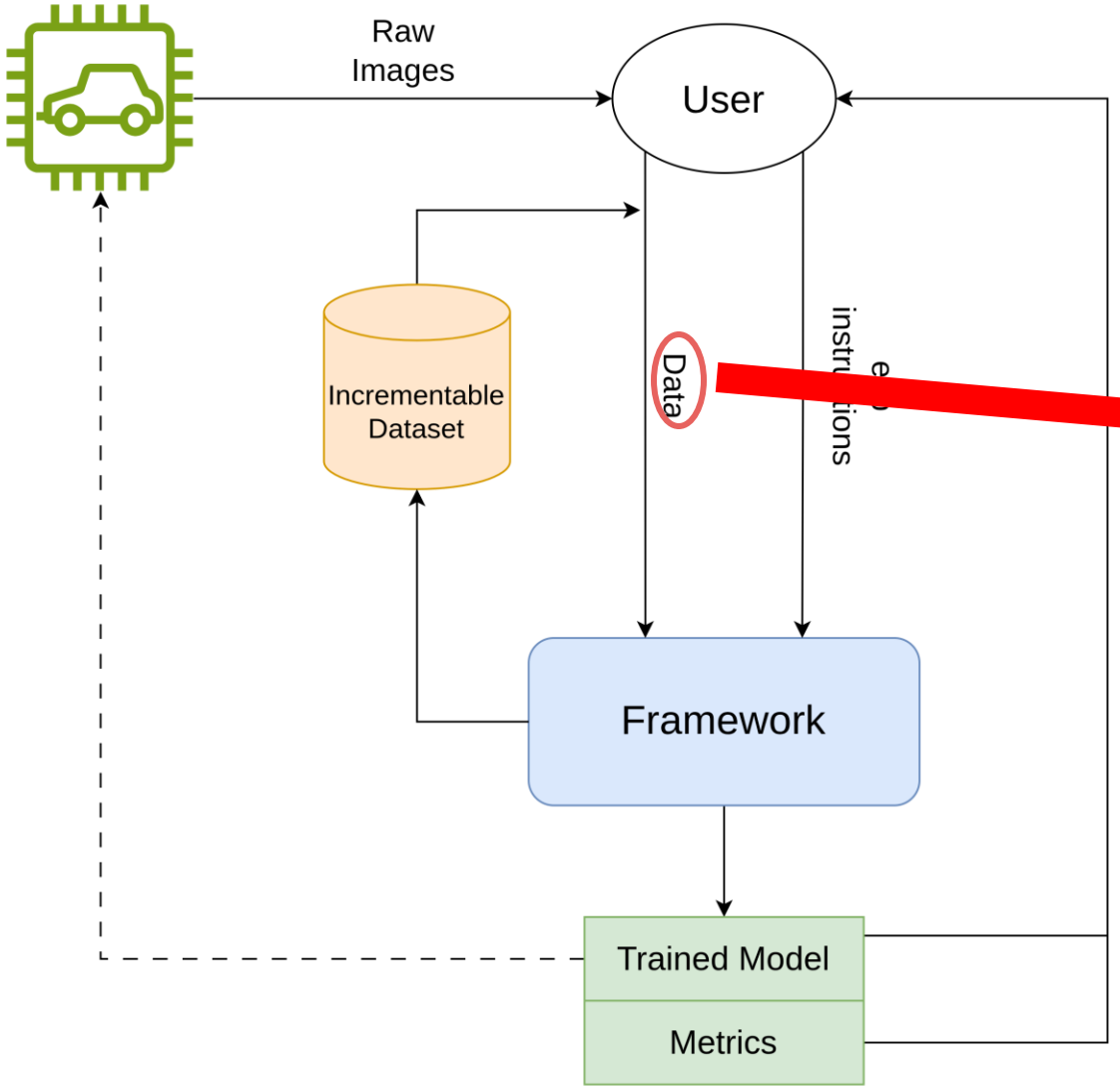
State of the Art Alternatives



Avalanche powered by ContinualAI

EDGE IMPULSE

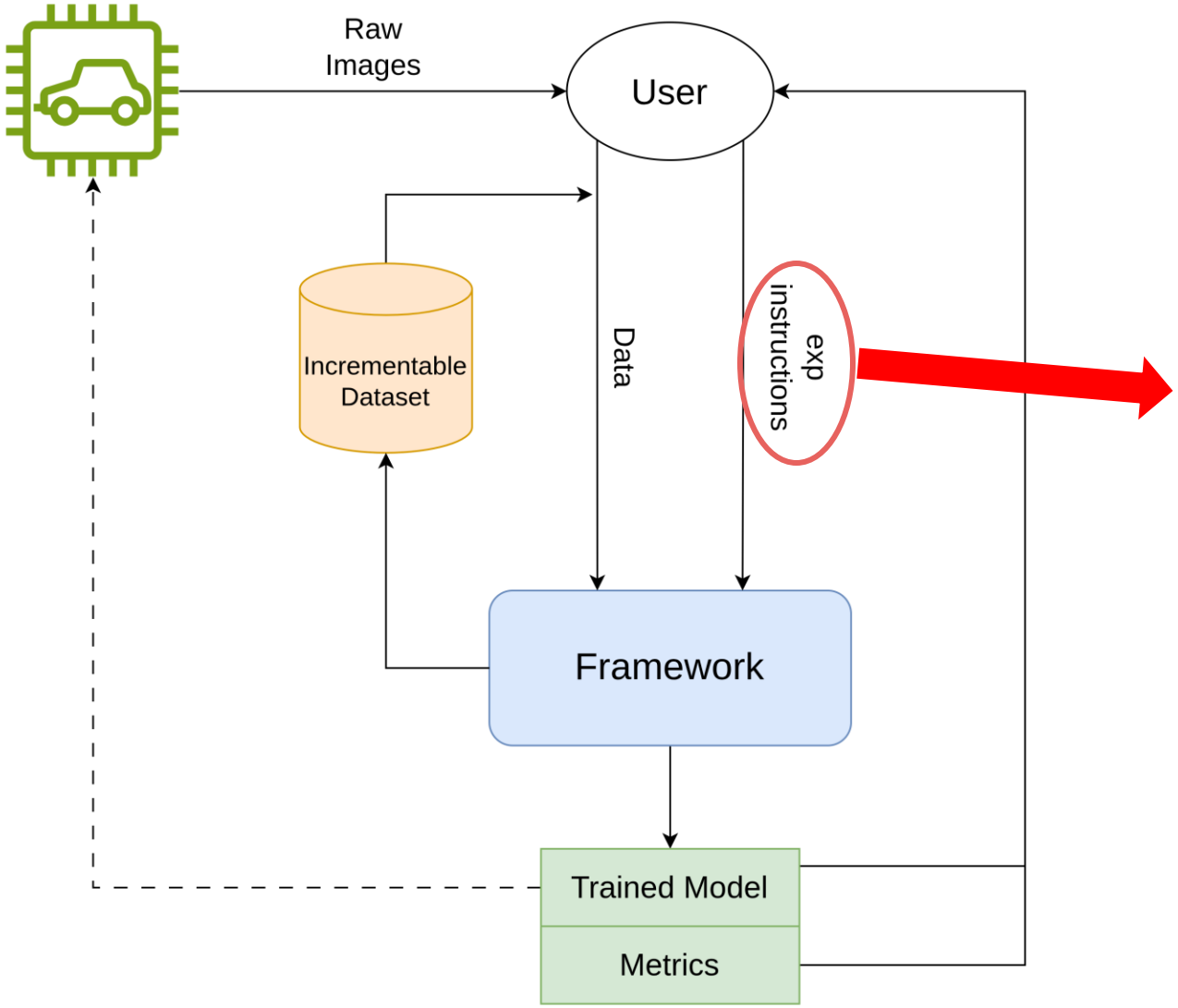
Entrypoint #1




```
datasets:
  dataset1:
    name: "ds_name"
    root: "/path/to/project/datasets/ds_name"
    type: "subfolders" # or "standard"
```

Descriptor for new session acquisition and storing

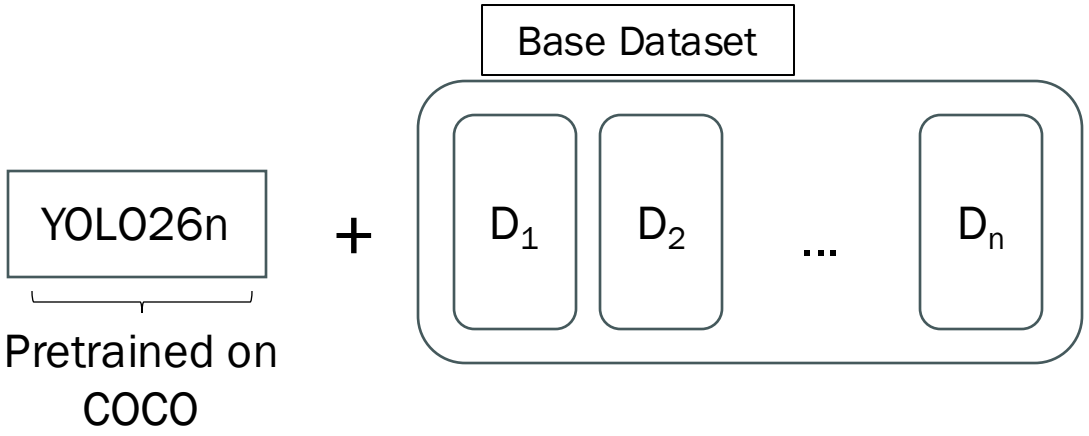
Entrypoint #2



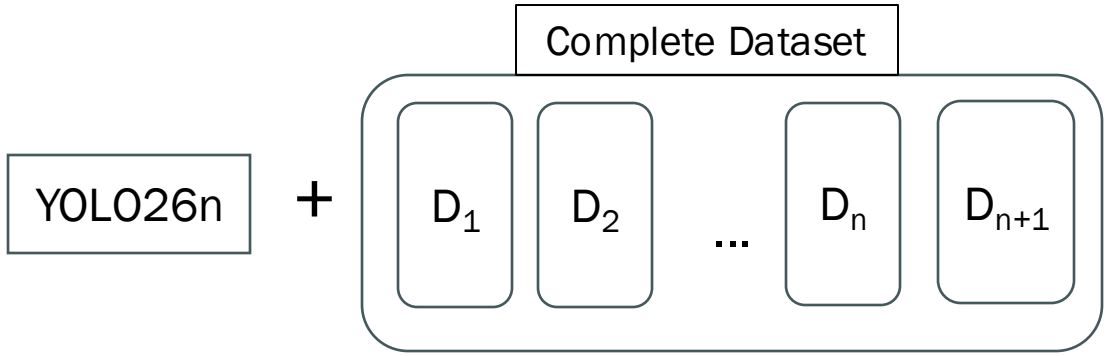

```yaml
experiments:
  exp1:
    name: base_90_10_s1
    model: yolo26n.pt
    hyp: {epochs: 100, seed: 1}
    datasets:
      squeaky_1Macchina:
      {pos_ratio: 1.0, neg_ratio: 0.1022}
      FIM_outside_base:
      {pos_ratio: 1.0, neg_ratio: 0.1022}
      ...
    val_pct: 0.15
    Seed: 42
```

Descriptor for new experiment addition

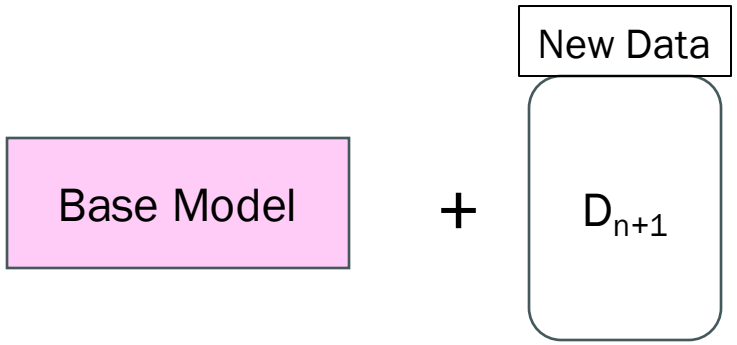
0. Base Training



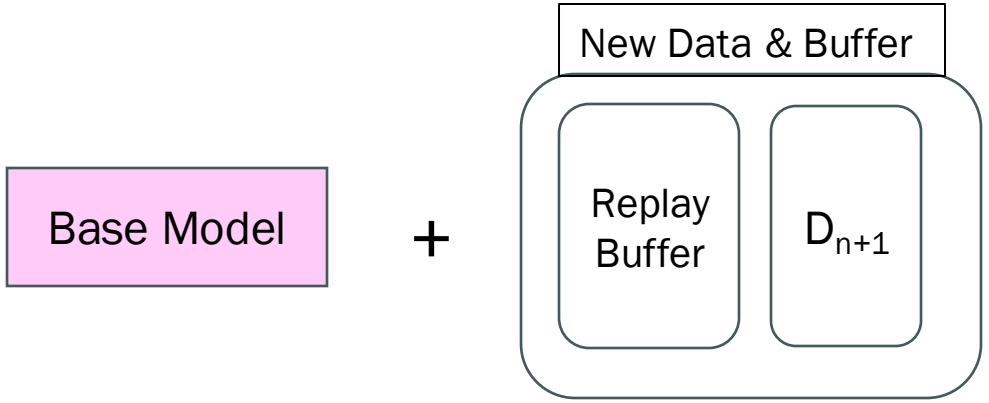
1. Joint Retrain



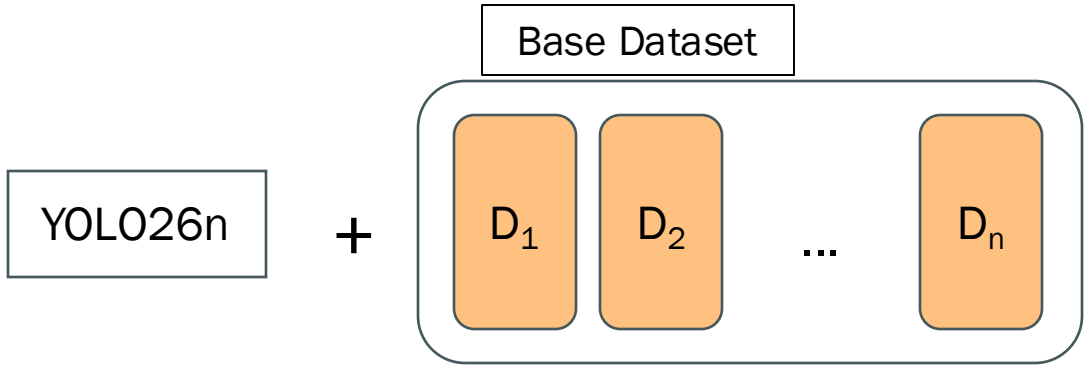
2. Fine Tuning



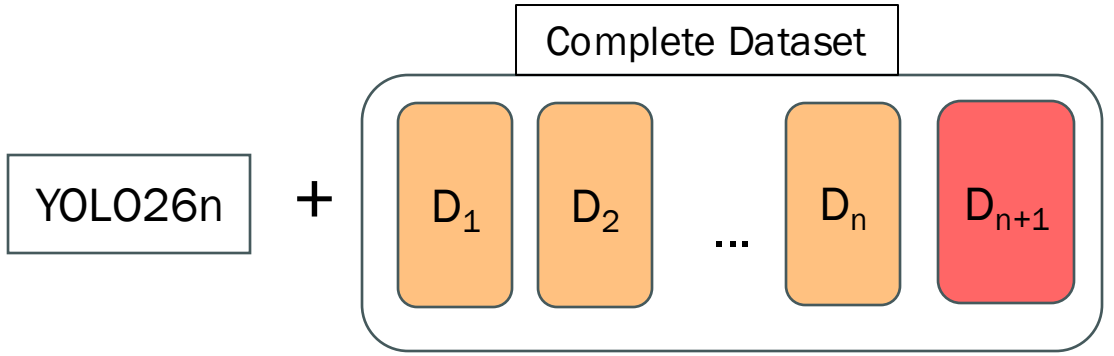
3. Continual Learning



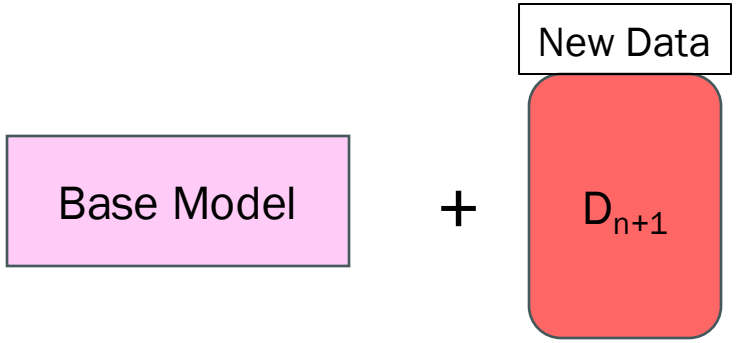
0. Base Training



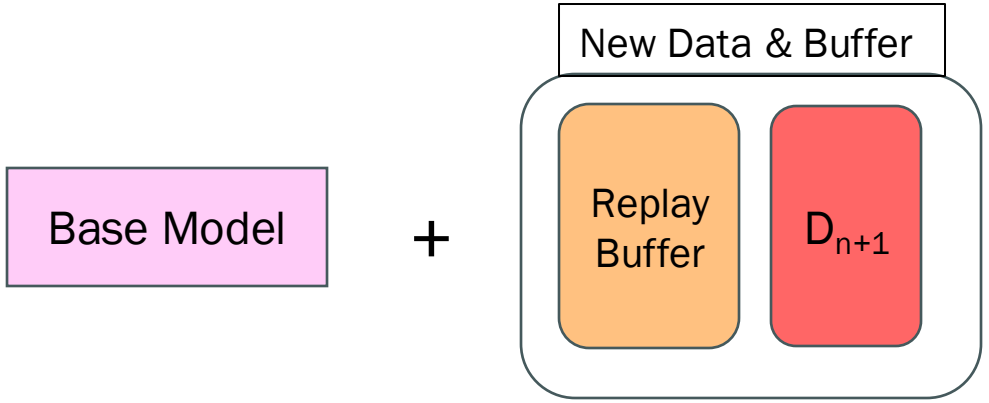
1. Joint Retrain



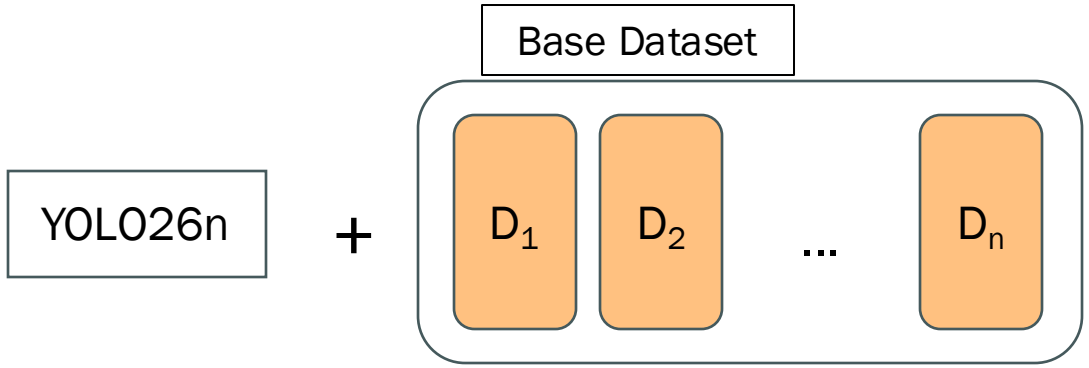
2. Fine Tuning



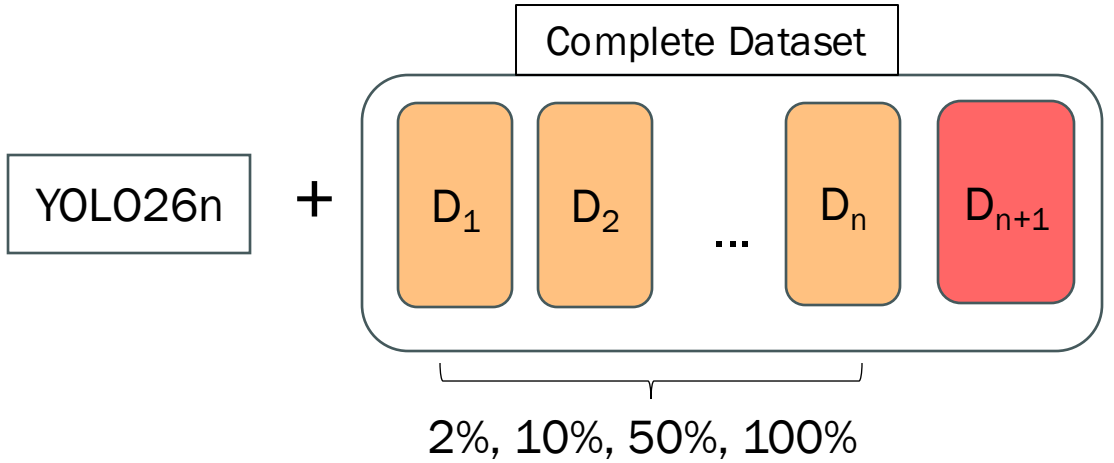
3. Continual Learning



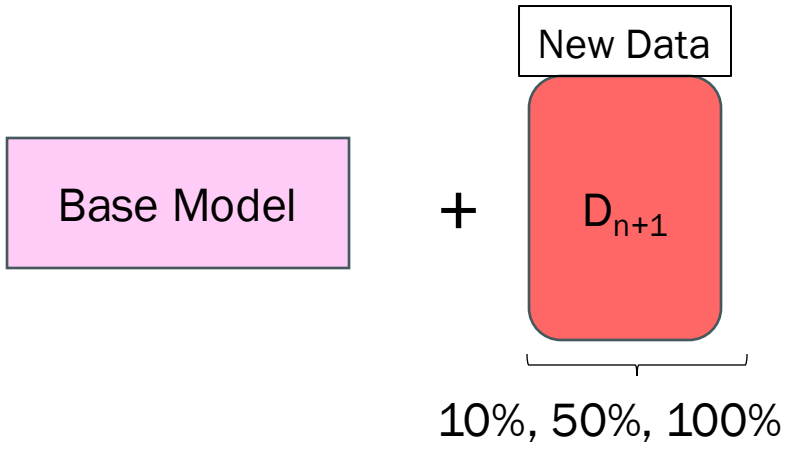
0. Base Training



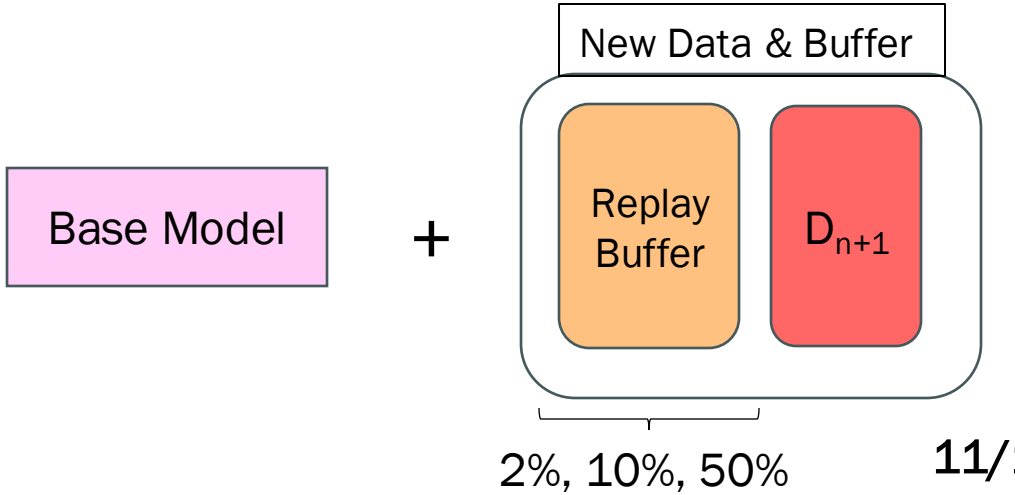
1. Joint Retrain



2. Fine Tuning



3. Continual Learning



	Dataset	Images	Positive	Negative
D_1	f1tenth.v2i.yolov11	2283	2283	0
D_2	huawei-uci2020-dataset	514	60	454
D_3	squeaky_17-04-26	3270	850	2420
D_4	objects_negatives_15-05-26	1165	0	1165
D_5	FIM_outside	3378	1932	1446
D_6	FIM_inside	2441	894	1547
D_7	icra_test_camera	3556	699	2857
	Total	**16607**	**6718**	**9889**

Already Available

Custom built

The Dataset In Numbers

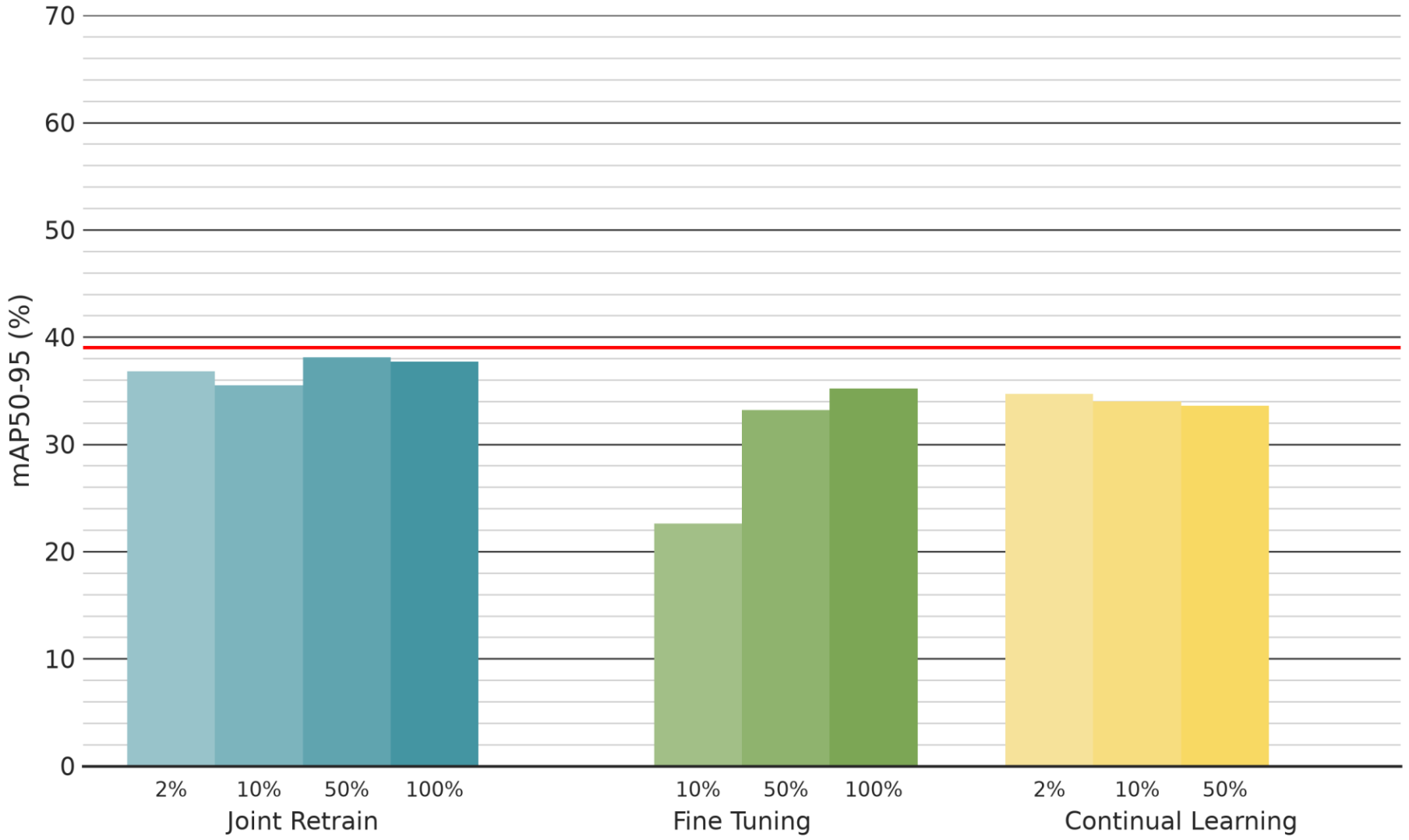
	Dataset	Images	Positive	Negative
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Specialization

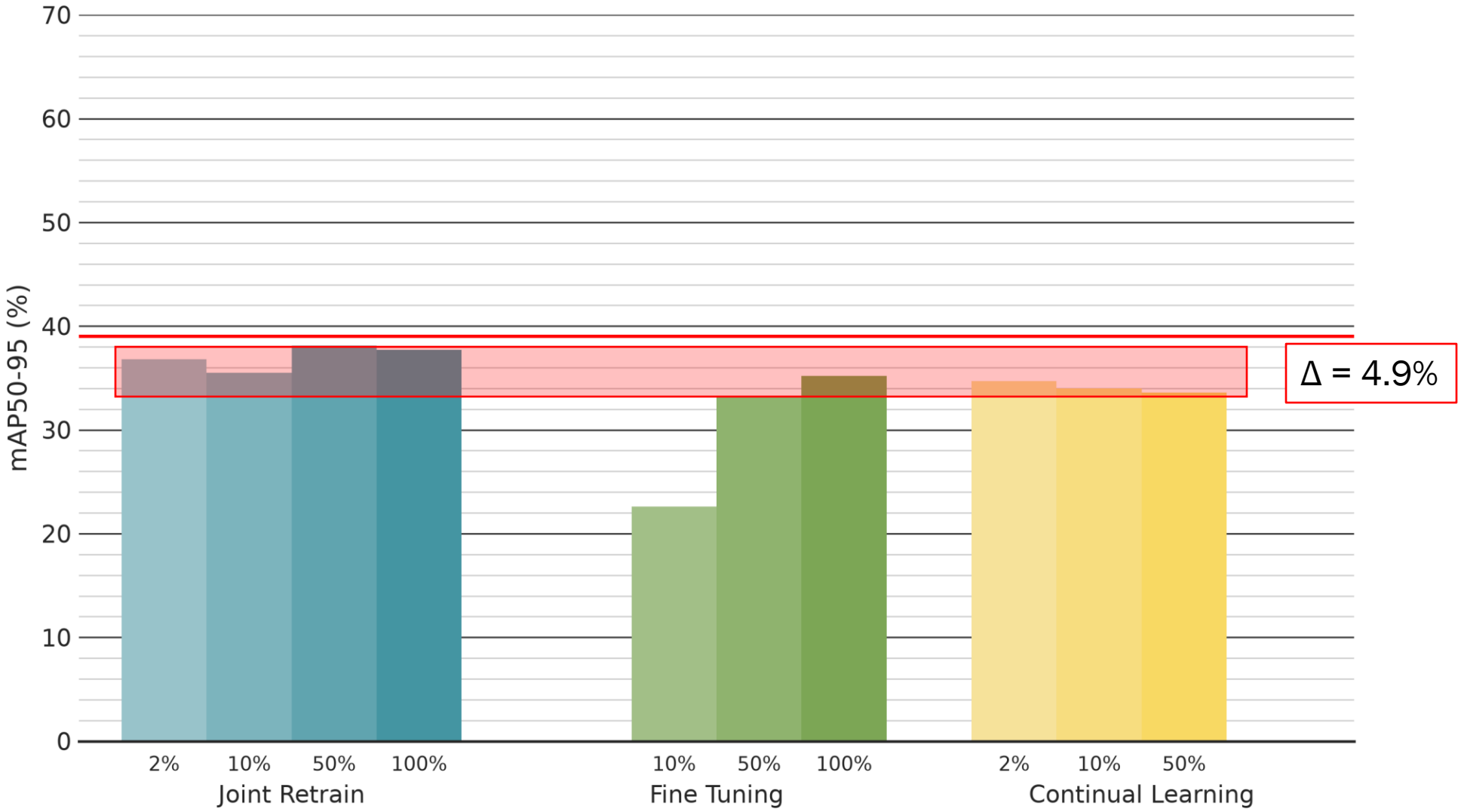
Forgetting

Generalization

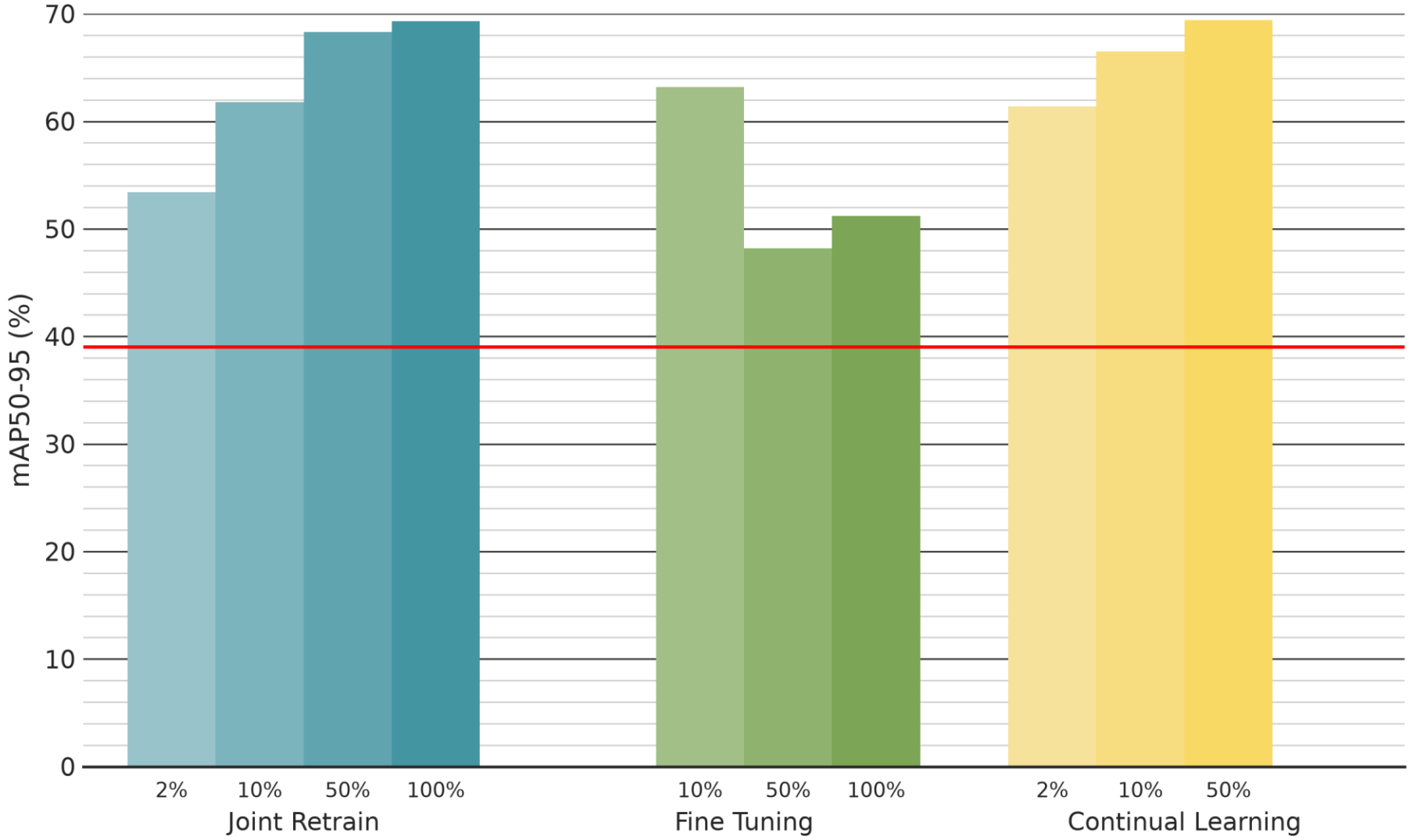
Test Sets & Experiments



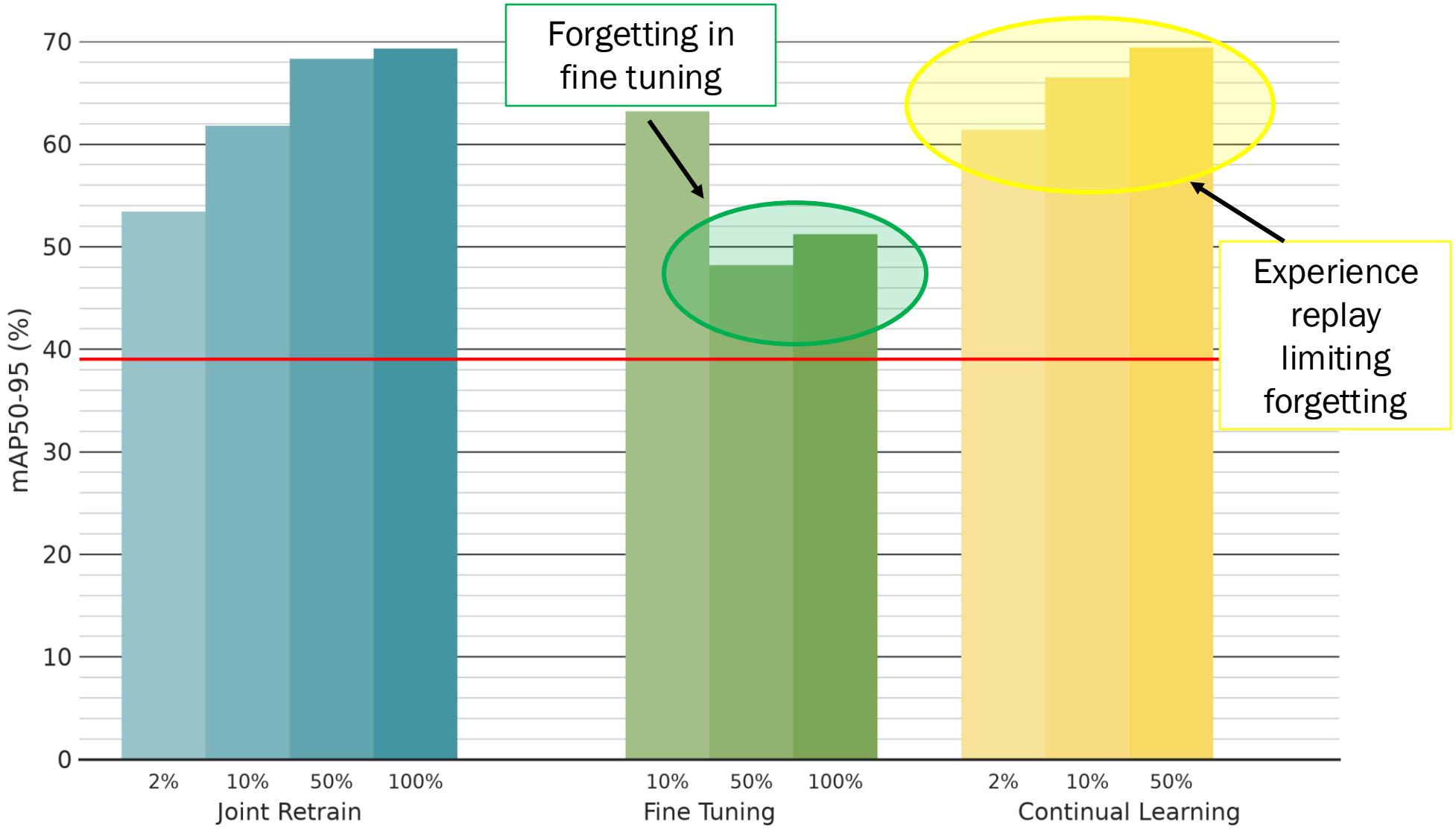
Specialization



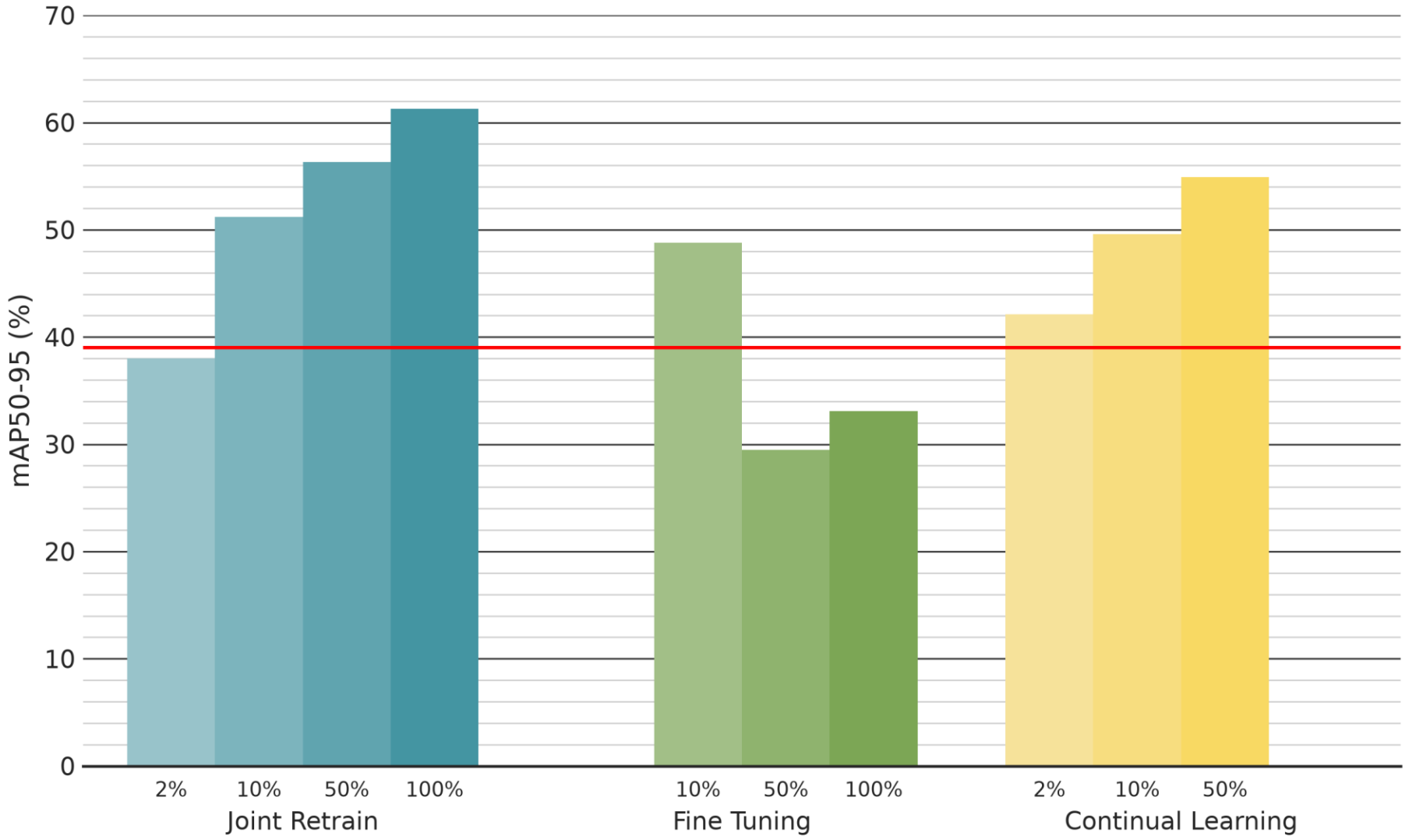
Specialization


70
60
50
40
30
20
10
0
mAP50-95 (%)
2%
10%
50%
100%
Joint Retrain
10%
50%
100%
Fine Tuning
2%
10%
50%
Continual Learning
YOLO26n

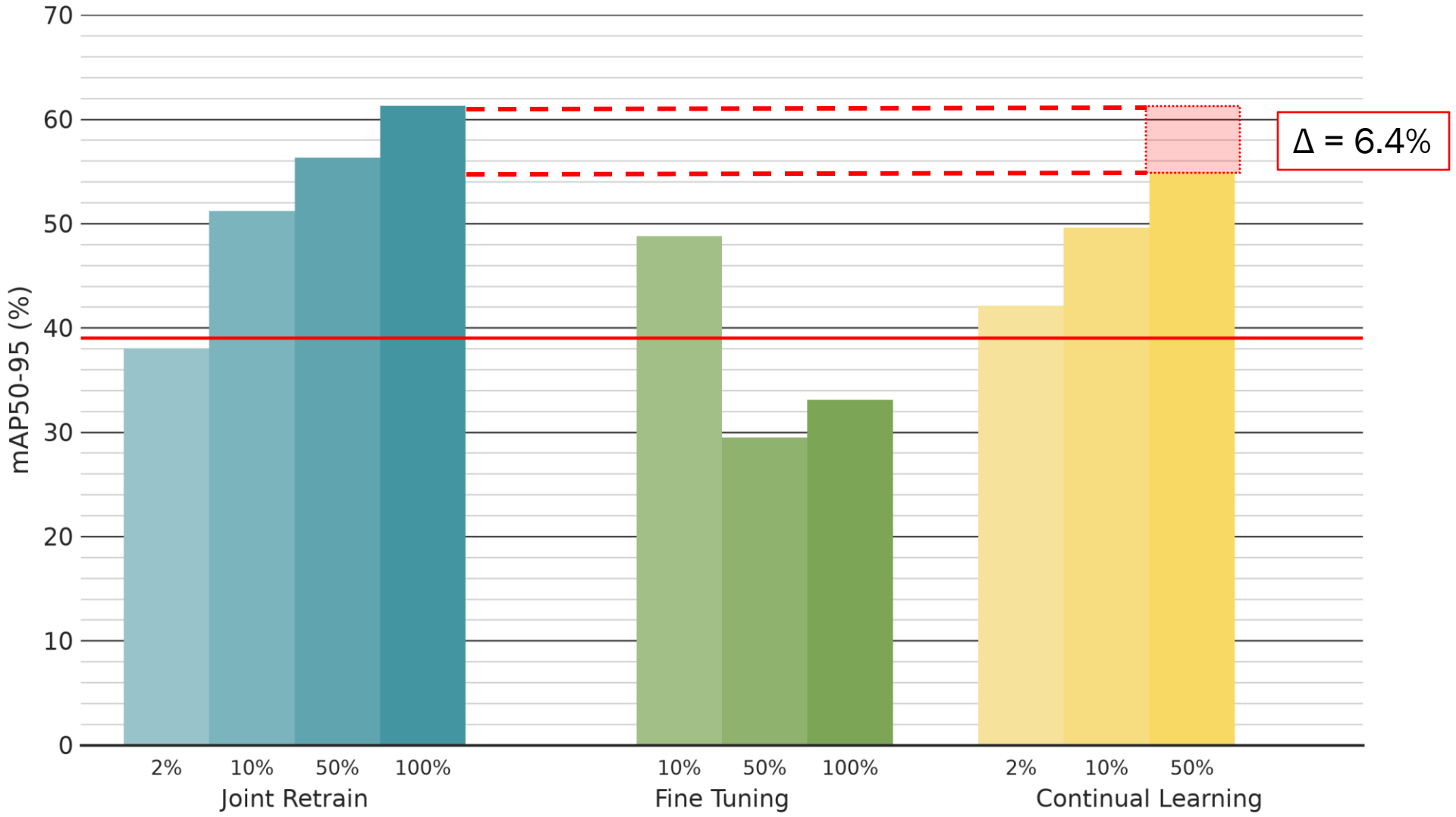
Forgetting


Forgetting in fine tuning
Experience replay limiting forgetting
mAP50-95 (%)
70
60
50
40
30
20
10
0
2%
10%
50%
100%
Joint Retrain
10%
50%
100%
Fine Tuning
2%
10%
50%
Continual Learning

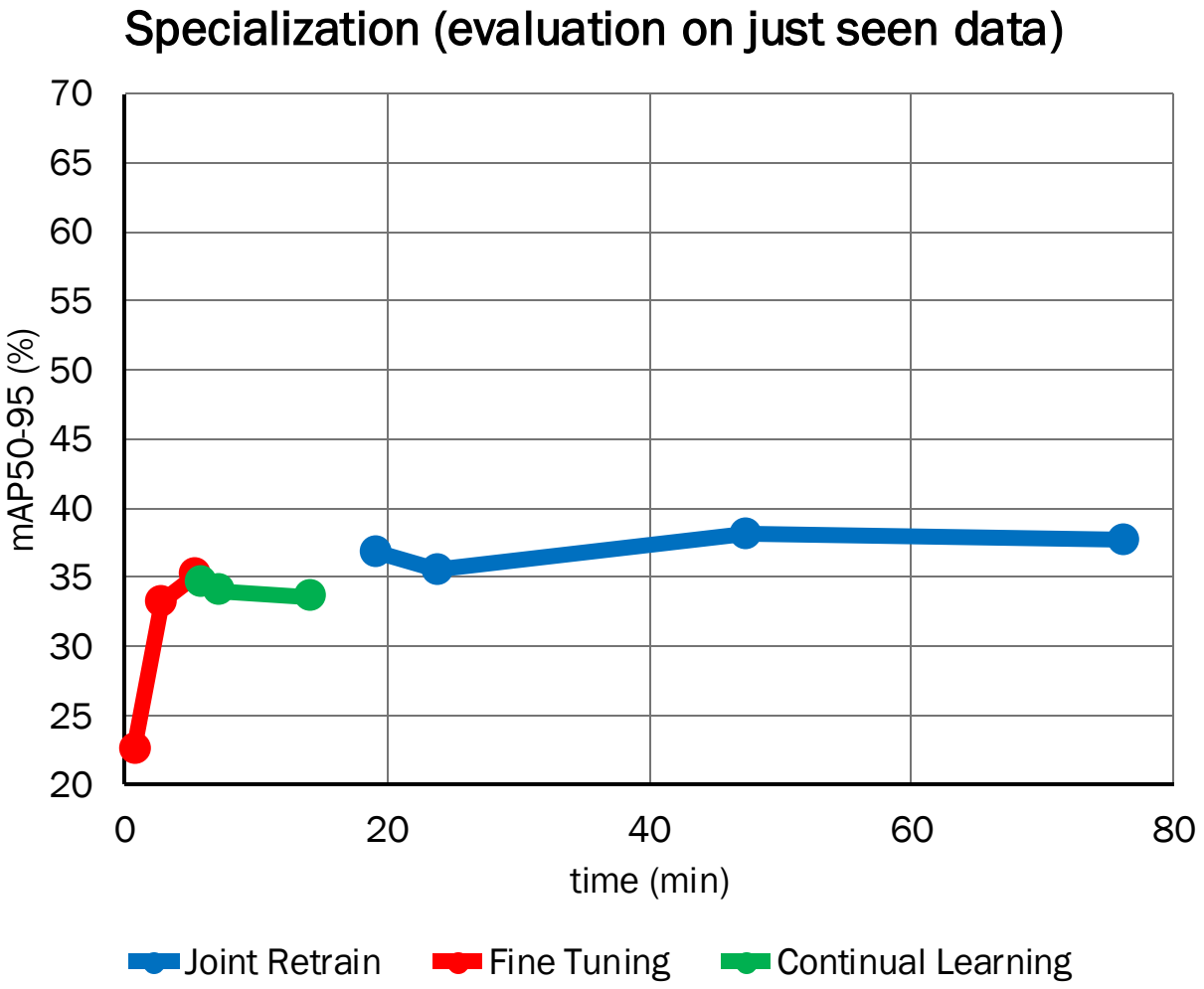
Forgetting

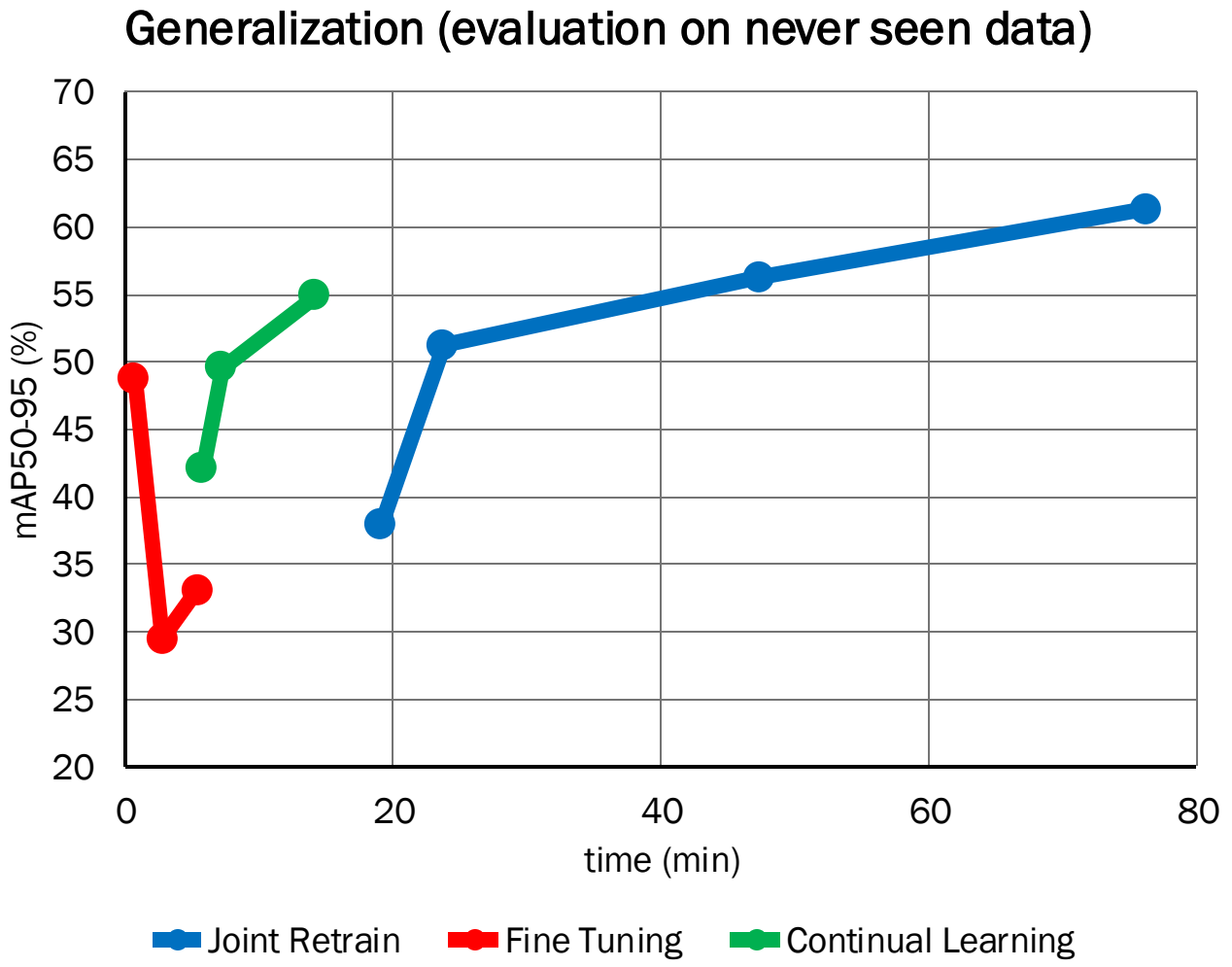


Generalization

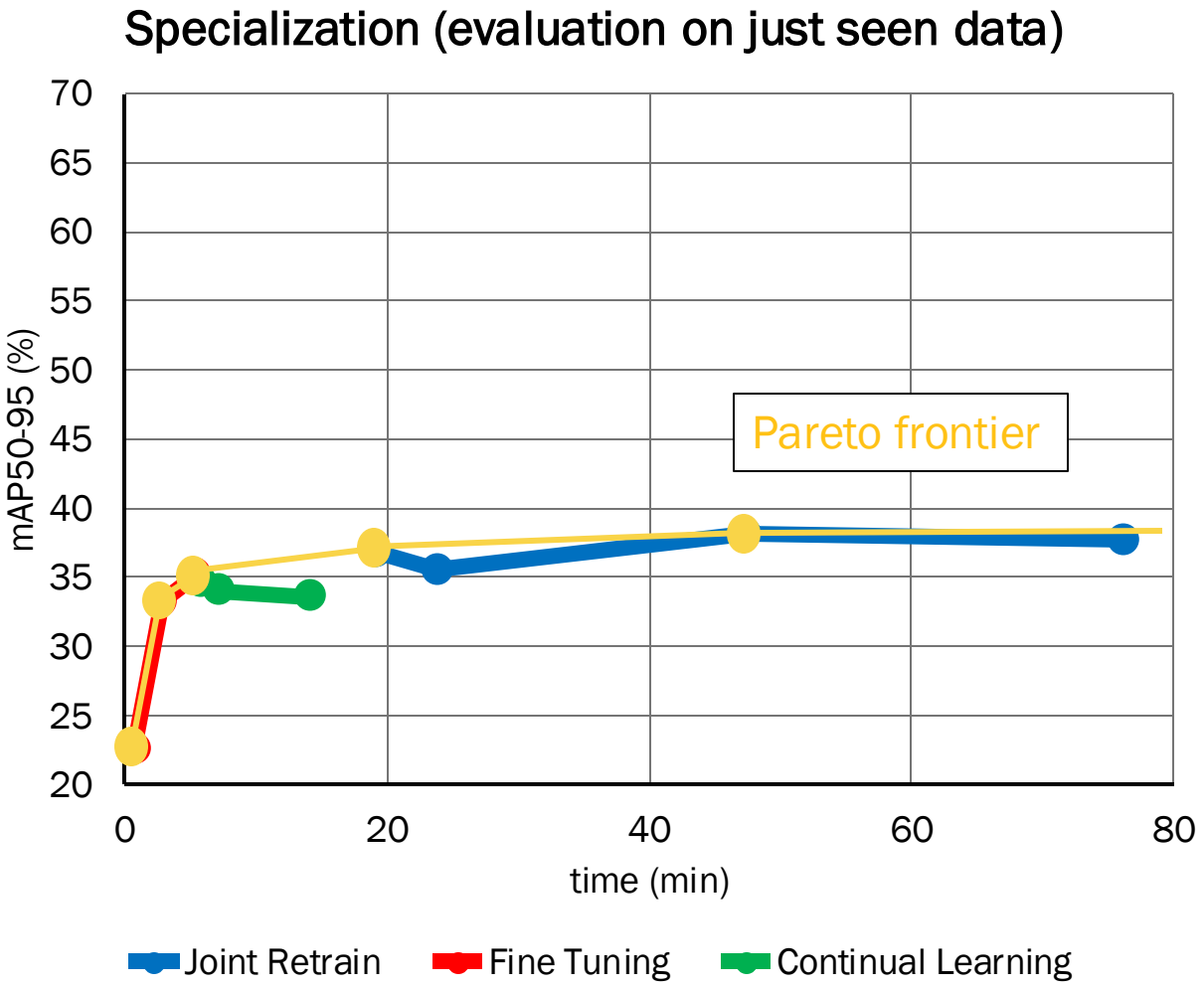


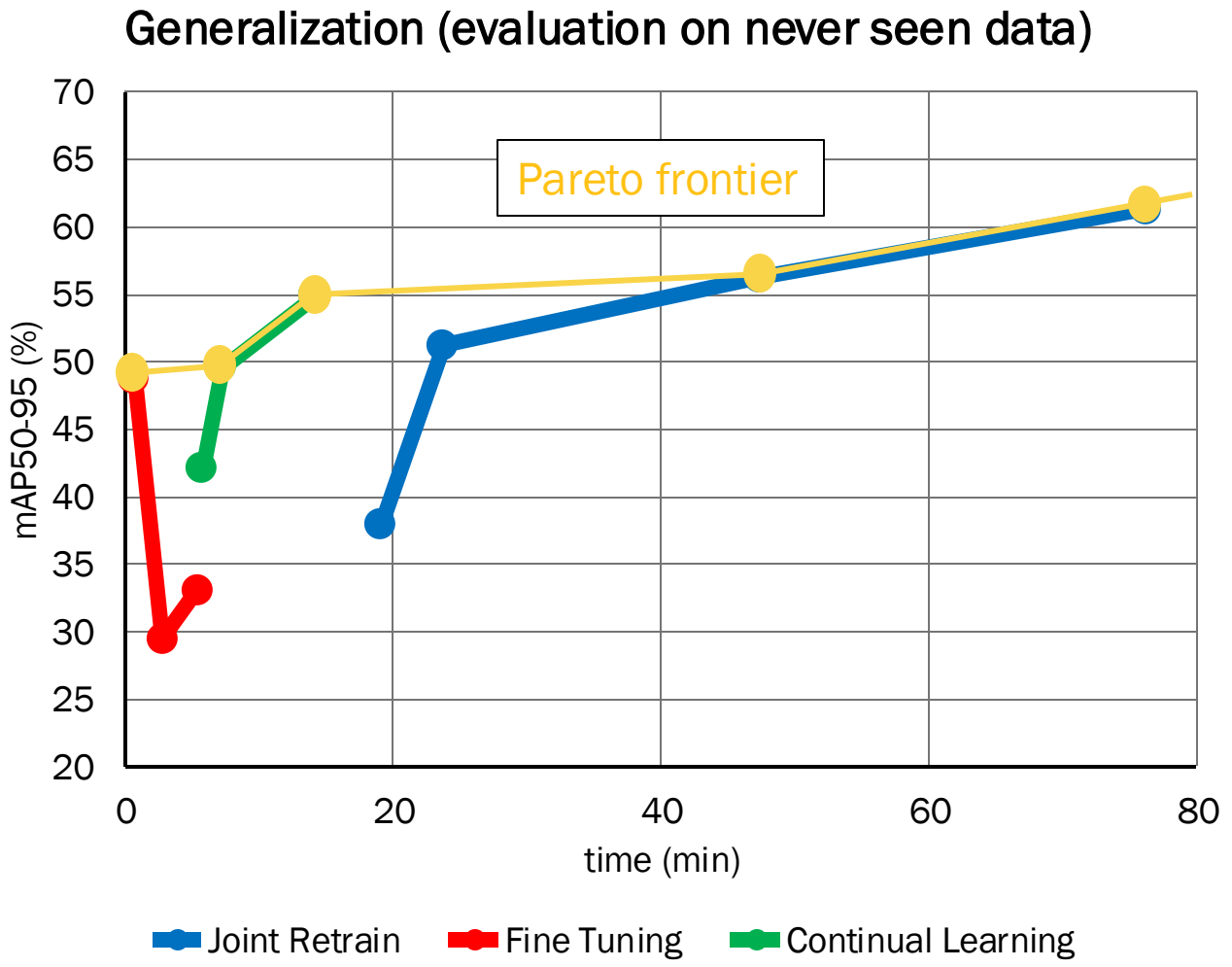
Generalization



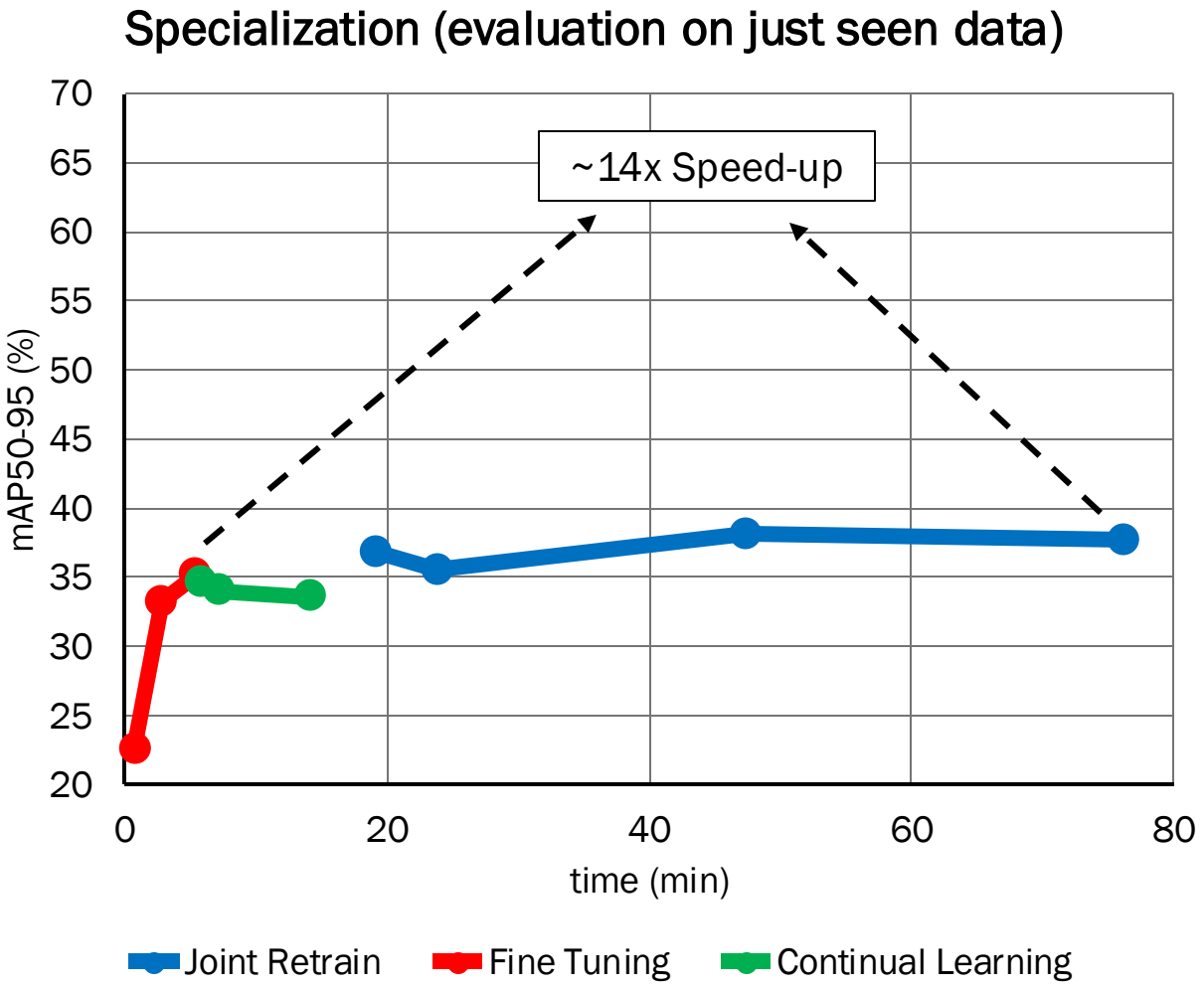


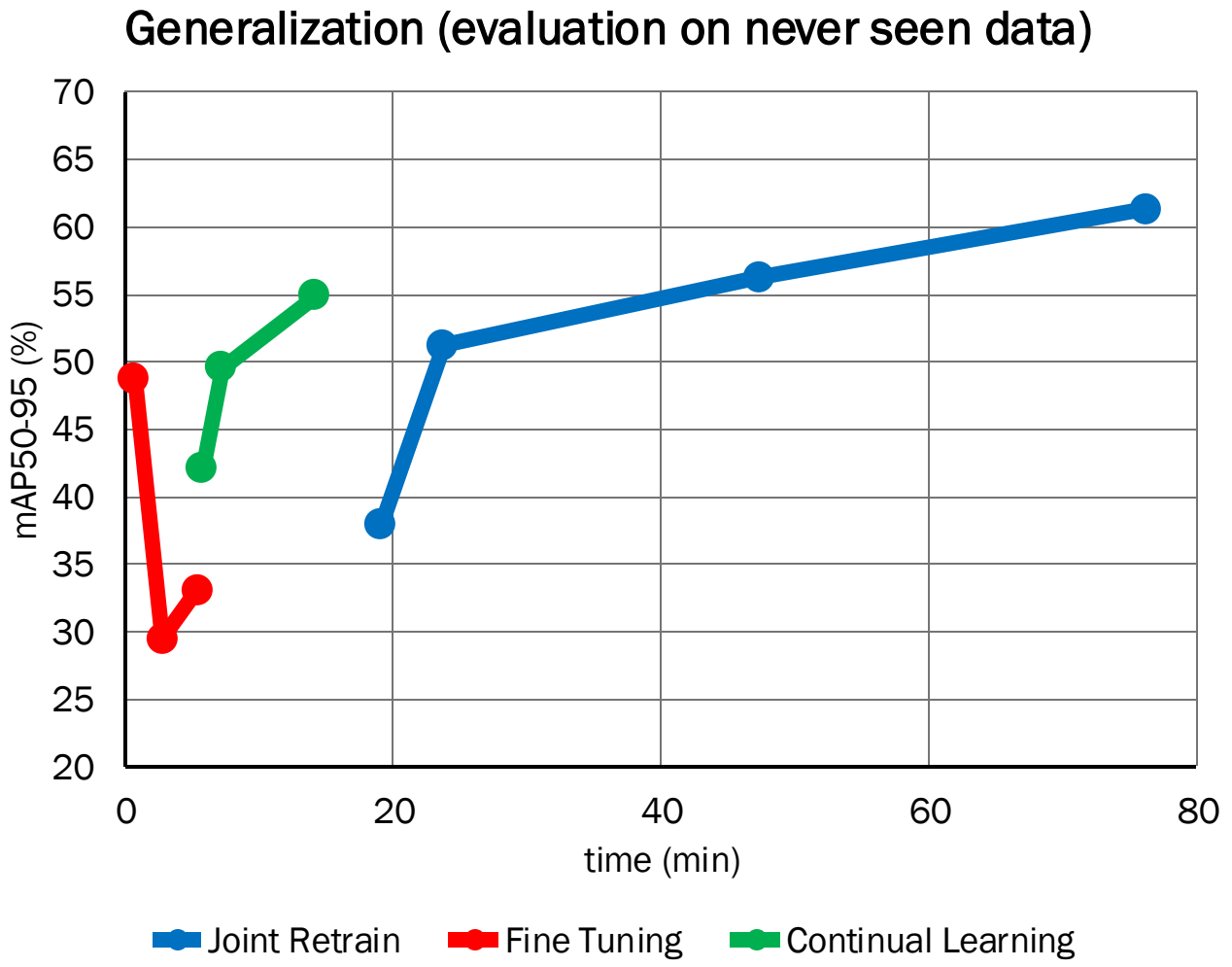
Training Time / Accuracy Tradeoff



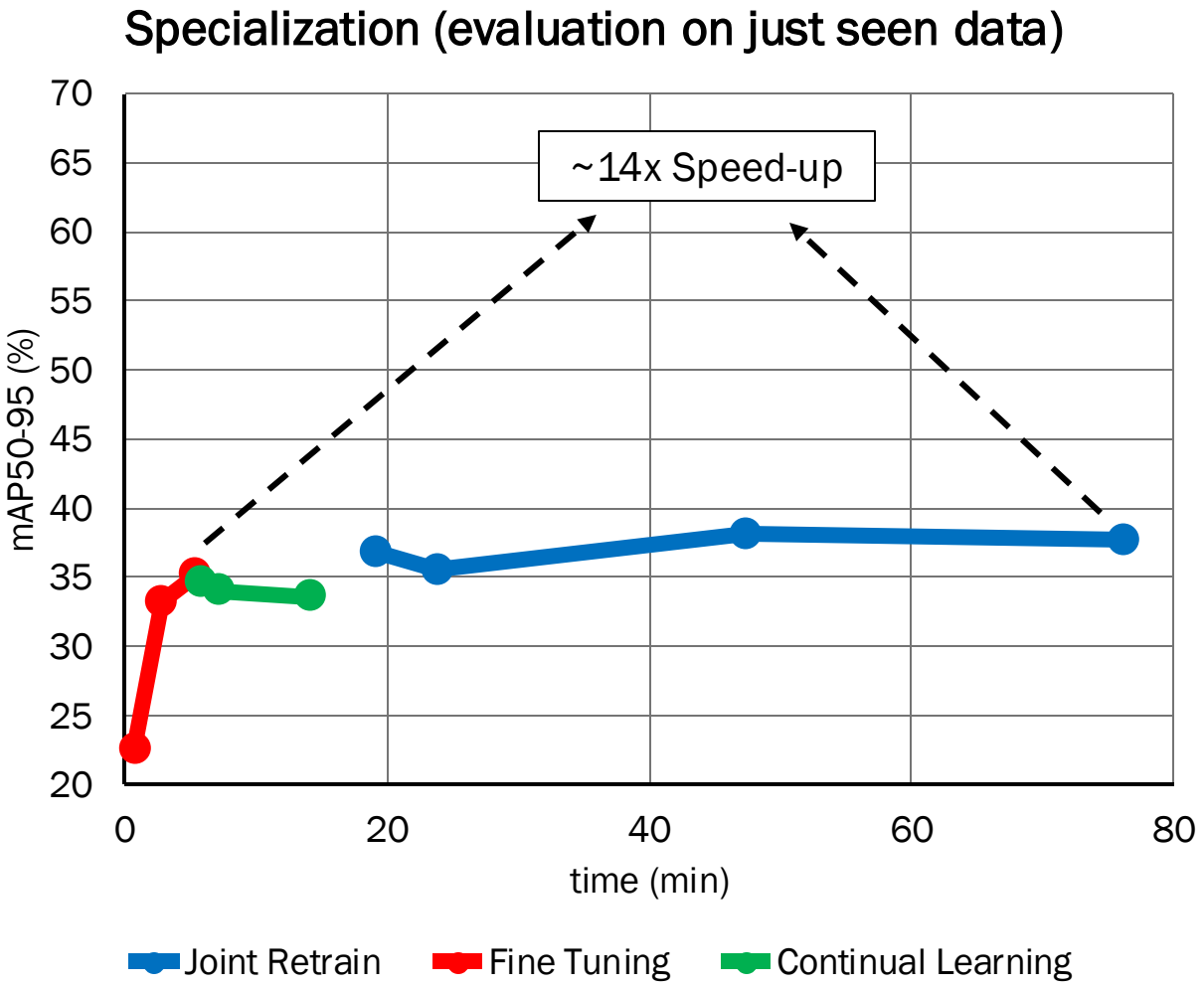


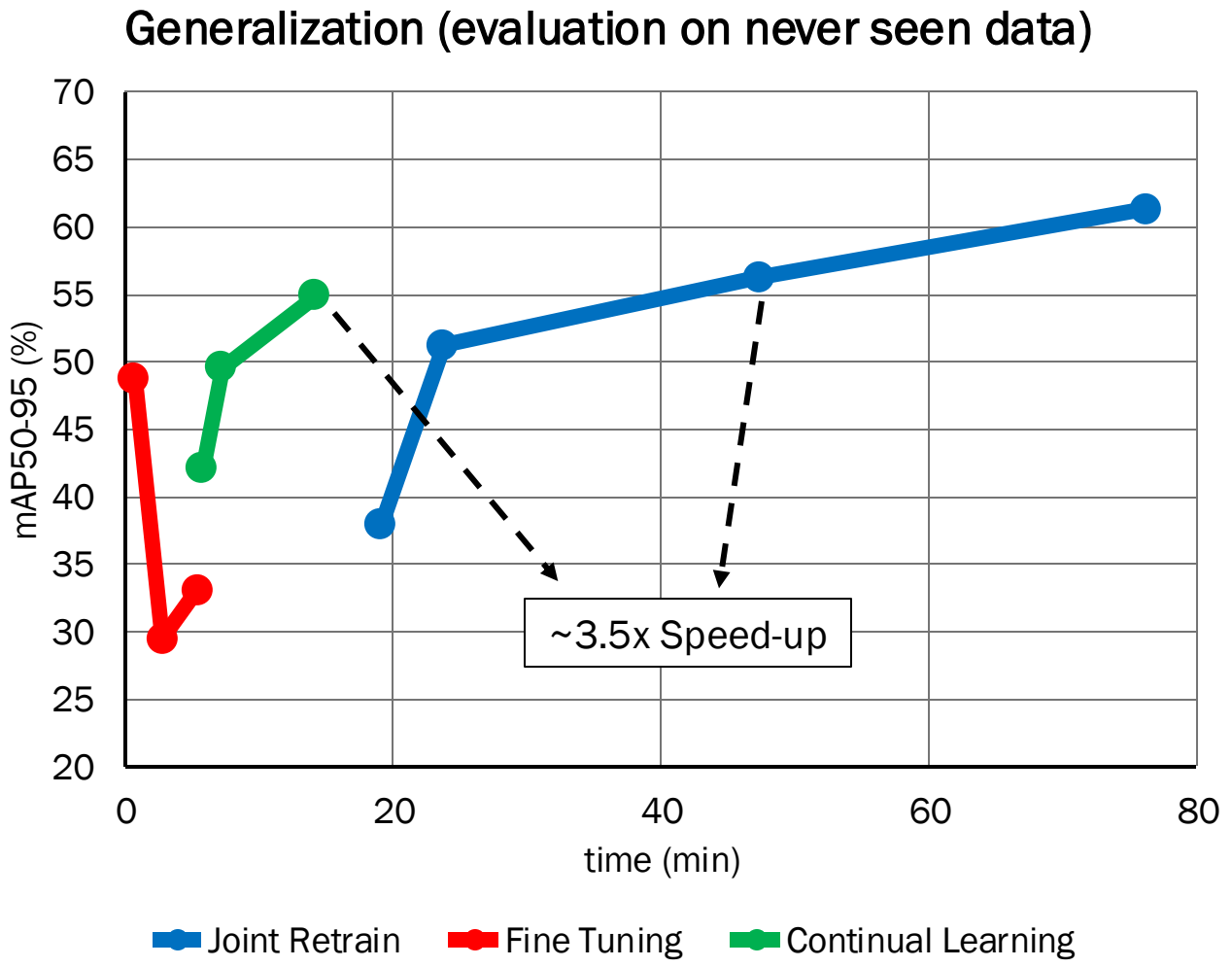
Training Time / Accuracy Tradeoff





Training Time / Accuracy Tradeoff





Training Time / Accuracy Tradeoff

Conclusions

Summary

1. Custom Incremental Dataset for RoboRacer
2. New Framework for Training
3. Joint Retraining, Fine Tuning and Continual Learning Experiments
4. Training Strategies with Results that Show Similar Performance with up-to 14x Speedup.
5. Possible Open Source Release

Future Works

1. Employment of Framework in Additional Use-Cases of the Framework: Unimore Racing and Generic Autonomous Racing.
2. Framework for Researching Different Models and Novel CL Methods
3. Research on On-Device Learning Strategies
4. Deployment on Edge Devices

That's all folks!

Tommaso Mortara

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Student ID: 206388